CJ Nour

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Professional Summary

Results-driven Electrical Engineer and recent McMaster B.Eng. graduate with over 3 years of professional experience at Husky Technologies, Tesla, and Moneris. Proven ability to deliver innovative electrical and automation solutions, specializing in industrial control systems, standardized designs, and cross-functional technical collaboration. Authored enterprise-level standards and drove engineering projects that sustained a multi-million dollar per year business. Excels in cross-functional environments and solving complex technical challenges.

Technical Skills

- Electrical Design
- EPLAN Electric P8
- Part Selection (UL/CE/IEC/NFPA 79)
- Power Distribution & Grounding
- Testing & Commissioning
- Computer Aided Design (CAD)
- Control Systems
- Beckhoff TwinCAT3
- Object-Oriented Programming

Professional Experience

Husky Technologies

Electrical Design Engineer — Jun. 2025 – Present

- Leading the electrical redesign of cabinets for a new machine generation. Creating schematics in **EPLAN**, CAD in **Siemens NX**, sizing cable assemblies, and validating grounding to **UL 508A**. Diagnosed and resolved a 9 year-old service issue related to Ethernet cable routing, **cutting critical communications faults by 100%**.
- **Resolved 30+ non-conformances** on a new machine family (UL & EN builds; CE and non-CE cabinets). Corrected panel schematic errors critical to safety devices, fixed assembly drawing errors. Achieved full release compliance by **eliminating 100% of the machine's electrical cabinets release blockers**.
- **Authoured and published two enterprise-level electrical standards** to drive proactive refresh of data and reduce engineering errors. Presented to engineering leadership to secure adoption across disciplines to actively **sustain a multi-million dollar per year business**.

Tesla

Manufacturing Controls Engineer Intern — Jan. 2024 – May 2024

- Spearheaded the development of an **RFID**-based monitoring system that aims to **reduce** tooling maintenance issues by **over 90%**. Integrated **PLC logic** and developed a custom **TwinCAT HMI dashboard** to track service history and collect tooling lifespan data on **multiple Gigafactory assembly lines**.
- Contributed to commissioning **3 automation** machines by leveraging **Beckhoff TwinCAT3 Structured Text** to program critical components, including **VFDs**, sensors, and robotic and pneumatic devices. Debugged field I/O networks using multiple industrial communication protocols (**TCP/IP, EtherCAT, Profinet**).
- Led multiple **R&D** initiatives to solve critical engineering challenges to enhance **Beckhoff PLC** and **FANUC robot** performance by experimenting with **computer vision systems**, **time-of-flight lasers**, **profilometers**. Authoured **test result reports** and applied technical specifications to optimize device integration.

Material Planner Intern — Sep. 2023 – Dec. 2023

- Developed and sustained 3 **Excel VBA** tools to optimize inventory and logistics, **eliminating duplicate shipments by 100%**. Automated the parts reordering process, automated gap analyses for project needs, and streamlined shipping with auto-generated forms and emails - enhancing database operational efficiency.
- Leveraged these tools to **identify inefficiencies** in the existing third-party inventory system and built a business case to influence executives to create an internal software team to develop an improved, proprietary system.
- Facilitated international shipments, expedited missing parts, and streamlined kitting to **reduce technician downtime by 20%**, ensuring efficient workflows across Gigafactories and optimizing daily resource allocation.

Professional Experience Continued

Moneris

Technology Governance Intern May 2023 – Aug. 2023

- Reduced the application database **by over 50%** through **cost center analysis** and a detailed **general ledger audit**. Retired outdated applications and underutilized software licenses, resulting in **significant cost savings**.
- Orchestrated collaboration across departments to gather, validate, and standardize data for **cloud migration planning**, ensuring a seamless transition phase and enhancing application scalability.
- Strengthened IT governance and compliance frameworks by aligning the application inventory with corporate standards to streamline software usage across teams, resulting in a **reduction of licensing costs by over 30%**.

Software Engineer Intern Jun. 2022 – Apr. 2023

- Developed a tool **boosting productivity for 30+ employees** by enabling credit card transaction analytics. Built with **JavaScript** and **React**, the tool allows users to customize dashboards and manage widgets seamlessly.
- Improved code efficiency by **50%** by implementing **Redux** and a backend system hosted on **Parse**, optimizing personal and team tools by implementing reducers, utilizing dispatch, and **querying user database** properties.
- Collaborated in **agile** feedback sessions with the Scrum Master, QA team, Architecture team, and other stakeholders to identify and resolve platform issues related to our proprietary payment platform testing tool.

Extracurriculars

General Motors EcoCar EV Challenge

Propulsion Controls and Modeling Team Member Sep. 2024 – Apr. 2025

- Developing **energy-efficient** and customer-focused control features as part of a competition sponsored by the **US Department of Energy** and **General Motors** involving 13 universities across North America.
- Modelling control systems in **MATLAB** and **Simulink** and deriving model requirements for a Cadillac Lyriq EV, ensuring precise alignment with vehicle specifications as outlined by GM and our team's performance goals.
- Testing and validating control system requirements using **Simulink**, ensuring compliance with benchmarks and functionality standards outlined by the competition organizers to achieve an **enhanced propulsion system**.

Projects

Autonomous Defense Drone System

- Developing an autonomous defense drone system to secure airspace by detecting, tracking, and intercepting unauthorized drones. The system integrates **multi-camera detection**, robotic arms designed in **CAD**, and an autonomous drone to provide a solution to drone-related threats.
- Utilizing **RF Tx/Rx** modules to enable communication between the camera hub system and the drones. Implementing **YOLOv8**, an **AI camera**, and **control algorithms** in **Python** to command the motors and autonomously track, pursue, and intercept unauthorized drones.

Autonomous RC Car

- Implemented self-driving algorithms coded in **C++ and Python** on an RC car to navigate through obstacles via **LiDAR** and **ultrasonic sensor** data processed through a **NVIDIA Jetson Nano** running on **Linux Ubuntu**.
- Coordinated **agile** team meetings as the Scrum Master to monitor progress, assign tasks, address challenges, and troubleshoot issues during development and testing phases.

Automated Vanity System

- Designed and fabricated a custom automated vanity system from a recycled oil barrel. Cut and welded a panel from the barrel to a recycled chair and added wheels for smooth accessibility.
- Engineered a motorized mirror and a **Bluetooth** sound system by wiring a recycled car window regulator motor to a momentary rocker switch and a power supply with an **AC/DC rectifier**. Researched **torque-speed characteristics** and motor specifications to ensure the motor could support the load with the supplied power.

Education

B. Eng., Electrical Engineering Expected Apr. 2025

McMaster University Hamilton, ON

- **Relevant Coursework:** Power Electronics, Electric Motor Drives, Control Systems, Intro to Mechatronics, Electromagnetics I & II, Electric Devices & Circuits, Microprocessor Systems, Signals & Systems, Logic Design
- **Special Interests:** Guitar, Music Production, 3D-Printing, Basketball, Skiing, Cycling, Running